

1. Explain Referential Integrity with Sample Queries

Referential integrity ensures that relationships between tables remain valid. In our experiment, inserting a student whose dept_id does not exist produced a foreign key error, proving that the database prevents invalid references. Example: INSERT INTO student VALUES (1005, 'Sam', 'Male', 101) failed because dept_id=101 did not exist in the Department table.

2. Explain JOIN in SQL with Sample Queries

JOIN combines rows from multiple tables based on shared keys. A LEFT JOIN returned three students even though no matching department existed, while a RIGHT JOIN returned one department (Physics) without students. A UNION of LEFT and RIGHT JOIN simulated a FULL JOIN and produced four results.

3. Compare Developer API vs User API

User APIs are designed for end users to access functions easily, such as weather or messaging apps. Developer APIs allow programmers to integrate systems and automate actions, such as REST endpoints or SDK libraries.

4. Explain Components of REST API

REST APIs communicate between client and server using standard rules. Components include Resources (objects), URIs (addresses defining resources), Methods (GET/POST/DELETE), Headers (metadata, authentication), and bodies (JSON data sent between systems).

5. Compare HTTP Methods

GET retrieves data without resource modification. POST creates new data. PUT replaces existing data fully. PATCH modifies only fields. DELETE removes resources. Correct method selection enforces proper API behavior.

6. Explain Authentication Field in HTTP Header

The Authorization header validates user identity before access is granted. Examples include Basic token strings or Bearer JWT tokens. The server checks this to prevent unauthorized access.

7. Explain Cookies Field in HTTP Header

Cookies store user session data so the server can remember state across multiple requests. A login may issue a cookie via Set-Cookie, and future browser requests return the cookie automatically.

8. Purpose of HTTP Response Headers

Response headers return metadata like content type, caching rules, and security policies. They are required so the client knows how to interpret and safely handle the response.